

Explicit finite-order lower bounds for high-order robust composite Z rotations

Dytchem¹

¹*Independent researcher**

(Dated: October 2026)

Abstract

We study the unavoidable cost of a high-order robust single-qubit Z rotation in a minimal model: the Z evolution is imperfect, its angle carrying a common unknown multiplicative error, while exact, instantaneous and free X rotations are available in unlimited number and at arbitrary positions. The resource is the total nominal Z angle T . We prove explicit lower bounds on T that hold at every order N of error cancellation, with all constants written out. They form a ladder of four inequalities with asymptotic coefficients $2/e$, $4/e$, $2\pi/e$ and 4 , obtained from a single Taylor step, from Jensen's formula for the carrier, and finally from the whole frequency band of that carrier. The last of them is an exact criterion: whenever an explicit closed-form quantity in a trial bandwidth τ falls below $\sin^2(\phi/4)$, where ϕ is the target angle, every order- N sequence satisfies $T > 2\tau$. It certifies $T > 2\tau$ with $2\tau = 2, 2, 4, 8, 10, 14, 16, 18, 22, 24, 28$ at $\phi = \pi$ for $N = 2, \dots, 12$, with all constants explicit and machine-checked. Its closed form gives the asymptotic coefficient 4 . We also delimit the method: whether any argument that uses only the bandwidth and the flatness order can pass 4 is equivalent to an explicit uncertainty-principle-type statement, the remaining gap is the interval $[4, 2\pi]$, and incoherent errors place the useful order near $N = 2$ – 4 . All bounds beyond the first are formalized in Lean 4 with Mathlib, as per-sequence statements.

I. INTRODUCTION

Composite-pulse methods suppress systematic control errors by replacing a single imperfect operation with a structured sequence whose error terms cancel order by order. Arbitrarily high-order cancellation was established in early work on composite pulses, and the optimal asymptotic scaling of the number of primitive pulses was later characterized for important pulse-length-error models [1–3]. A separate line of work has addressed lower bounds on the physical operation time of robust composite gates, including geometric bounds for first-order pulse-length-error robustness [4] and time lower bounds for first-order robustness under bounded control amplitude [5, 6]. Order- N cancellation is the vanishing of the first N error moments, the condition that also underlies optimized dynamical decoupling and the filter-function description of open-loop control [7–9], where rigorous design bounds of a

* Corresponding author: contact through the public project page <https://github.com/Dytchem/robust-z-lower-bound-lean>.

different type are available [10–12]. Section II compares these formulations with the one used here.

The resource model considered here is deliberately different. Suppose that one available interaction generates a Z rotation but its strength is not known exactly, and that a second control can rotate the system about an orthogonal X axis with sufficiently high precision that, in the idealized limit, its duration may be regarded as negligible. The natural question is then not how many control pulses are needed, but how much of the imperfect interaction must unavoidably be accumulated in order to cancel the common systematic error to a specified order. The distinction matters: a lower bound on the number of primitive rotations does not immediately give a lower bound on a time resource when some rotations are assigned zero cost.

The setting also occurs naturally in Rydberg-atom control. In a recent proposal based on quantum signal processing (QSP) [13], the interaction between two Rydberg atoms is treated as an unknown but deterministic phase input, while fast Rydberg transitions are used as interleaved controls, and systematic variation of the interaction strength is suppressed by composite control. Systematic control deviations are, independently of that proposal, among the dominant error sources of neutral-atom two-qubit gates [14, 15]. The present work isolates the single-qubit mathematical core of this setting: what is the unavoidable amount of an uncertain Z evolution if exact transverse rotations are free?

The results proved here are explicit, finite-order statements: lower bounds on T that hold at every prescribed order N , with all constants written out. They form a ladder of four rungs, in the order $2/e \rightarrow 4/e \rightarrow 2\pi/e \rightarrow 4$, each using strictly more of the structure of the problem than the previous one. Differentiating the propagator itself and taking one Taylor–Lagrange step at the endpoint gives $T \geq 2[2(N+1)! \sin(\phi/4)]^{1/(N+1)}$, a coefficient $2/e$ in N ; this rung uses nothing but the product rule, and it is kept to show how little is needed for linear growth. Doubling the flatness through a scalar carrier turns the same step into $T \geq 2[2q! \sin^2(\phi/4)]^{1/q}$ with $q = 2N + 2$, a coefficient $4/e \approx 1.4715$ (Theorem 3). The carrier extends to an entire function of exponential type $T/2$, so Jensen’s formula converts the same q -fold zero at the nominal point into $T \geq \pi q/[e(2/s)^{1/q}]$ with $s = 2\sin^2(\phi/4)$, a coefficient $2\pi/e \approx 2.3115$ with no threshold and no approximation theory (Theorem 4). Finally, the vanishing derivatives become q vanishing moments, into which any polynomial of degree $q - 1$ may be inserted; approximating $e^{-i\mu}$ on the whole band and smoothing

with a compactly supported Fourier multiplier makes the endpoint value exponentially small whenever an explicit closed-form criterion in the bandwidth is met, which contradicts the order-one separation of the endpoint and yields a checkable certificate $T > 2\tau$ at every order (Theorem 5), together with a Lean-certified finite- N table and a closed form whose coefficient approaches 4. The four bounds combine into a single bound valid at every order (Corollary 6), from which $\liminf_{N \rightarrow \infty} T_{\min}(N, \phi)/N \geq 4$ follows.

The control model, admissibility, the achievable costs, the carrier, and the three rungs beyond the first are contained in public Lean 4 + Mathlib developments [16]; the paper follows those developments closely, and Sec. VIII lists what is machine-checked, statement by statement.

The paper is organized as follows. Section II reviews the relation to previous work. Section III states the model and all four bounds up front, with no proofs. Sections IV–VI prove the rungs in ladder order, one rung family per section, with the shared carrier setup factored once into Sec. IV A and cited thereafter. Section VII reports numerical sequences as upper bounds only, with no optimality claim. Section VIII tabulates the correspondence with the formal development, and Section IX records what cannot improve the ladder, the open ceiling dichotomy at 4, and the orders that matter for a device. The appendices collect the closed forms and the kernel estimate behind the exact criterion, and the conventions of the numerical data.

II. RELATION TO PREVIOUS WORK

The earliest composite-pulse constructions showed that systematic pulse-length errors can be cancelled to arbitrarily high order [1], and Ref. [2] reviews the vocabulary of order- N robustness and of cost accounting in that setting. Low, Yoder, and Chuang subsequently proved that, in a standard primitive-pulse model, the number of primitive rotations required for arbitrary accuracy has optimal linear scaling in the robustness order, with constructions saturating that scaling [3]. Their statement has the same shape as Theorem 5—a linear lower bound in the robustness order together with a construction that saturates it—but the resource there is the number of primitive rotations, in a model where those rotations are themselves part of the costly control sequence. It therefore says nothing about the minimum Z -evolution time in the free- X model, which is the resource bounded here.

The identification of order- N error cancellation with the vanishing of the first N moments of the error is standard in dynamical decoupling and in the filter-function description of open-loop control: it is the defining condition of Uhrig’s optimized sequences [8], and it was put on a general footing by the transfer-function analysis of Ref. [9]. The composite-pulse counterpart, a single-qubit sequence analyzed order by order in its systematic pulse-length error, is treated in Ref. [7]. Section VI uses the same identity, but the carrier there is a function of a Z -evolution budget rather than of a pulse train, and the constant 4 comes from the fact that this carrier is band-limited.

Kukita, Kiya, and Kondo derived a lower bound on the operation time of first-order pulse-length-error robust composite gates from a geometric argument [4]. In their resource model the rotations that change the control axis cost time; here they are instantaneous and free, so a geometric bound on the full control path is not a bound on T .

Two recent works bound noise-robust control from the side of the control amplitude. Zeng and Deng prove a universal average-case time lower bound for first-order robustness against a single quasistatic coherent error under bounded control amplitude, together with a construction that is robust in time $4T$ plus a constant [5]; that overhead is fixed and N -independent, whereas the coefficient studied here is linear in the robustness order. Kosut, Lidar, and Rabitz prove a universal lower bound on the fidelity of any robustly controlled gate as a function of the gate duration times an aggregate uncertainty [6]. Both concern first-order or order-independent robustness, and neither is sharp in the way the rungs are for the model of Sec. III: the present proof uses the analytic dependence on a common scale parameter λ and holds for arbitrary order N .

Machine-checked quantum information is by now a developed subject—quantum programs and compilers in Isabelle/HOL [17], and quantum information theory in Lean [18]—but we are not aware of a formalized lower bound in quantum control. The developments behind Theorems 3, 4 and 5 attempt one, for a model small enough that the whole proof, analytic estimates included, is checked by the kernel.

The connection to the unpublished Rydberg QSP proposal [13] is complementary. That work treats an unknown interaction phase as the input to a QSP-style composite protocol and uses fast control pulses to suppress systematic interaction variation. The present result identifies a resource limitation in the idealized single-axis core of that philosophy: even with perfect free transverse controls, high-order cancellation cannot be made sublinear in the

amount of uncertain evolution.

III. CONTROL MODEL AND MAIN RESULTS

A. Elementary rotations and composite sequences

Let

$$X(\alpha) := \exp\left(-i\frac{\alpha}{2}\sigma_x\right), \quad Z(\theta) := \exp\left(-i\frac{\theta}{2}\sigma_z\right), \quad (1)$$

where σ_x and σ_z are the Pauli matrices. Throughout, $\|\cdot\|$ is the operator norm on matrices (the largest singular value), which equals one on unitary matrices, and all numerics reported in Sec. VII use the same norm. The X rotations are assumed exact, instantaneous, and free. The Z evolution is the only costly operation, and in the presence of a fixed unknown fractional error ε its actual propagator is

$$Z^\varepsilon(\theta) = \exp\left[-i\frac{(1+\varepsilon)\theta}{2}\sigma_z\right]. \quad (2)$$

For $L \geq 0$, choose real angles α_j and nonnegative Z angles θ_j . Following the recursive order used in the Lean development, define

$$U_\lambda = X(\alpha_1)Z(\lambda\theta_1)X(\alpha_2)Z(\lambda\theta_2)\cdots X(\alpha_L)Z(\lambda\theta_L), \quad (3)$$

with $U_\lambda = I$ for $L = 0$. Relabeling the indices gives the equivalent convention in which $X(\alpha_L)Z(\lambda\theta_L)\cdots X(\alpha_1)Z(\lambda\theta_1)$ is written from left to right. Nothing in the proof depends on this notational choice.

The parameter λ is an interpolation variable: the physical propagator at error ε is $U_{1+\varepsilon}$. The target is

$$U_1 = Z(\phi), \quad 0 < \phi \leq \pi. \quad (4)$$

The cost is the total nominal Z angle,

$$T := \sum_{j=1}^L \theta_j. \quad (5)$$

The normalization is chosen so that the nominal Z rotation rate is one; restoring a physical rate ω simply replaces T by T/ω in the time interpretation.

Definition 1 (Order- N robustness). A sequence is *order- N admissible* for ϕ if $\theta_j \geq 0$, $U_1 = Z(\phi)$, and

$$\left. \frac{d^k U_\lambda}{d\lambda^k} \right|_{\lambda=1} = 0, \quad k = 1, \dots, N. \quad (6)$$

This is exactly matrix-valued flatness at the nominal error point.

Let $\text{costs}(N, \phi)$ denote the set of costs of all admissible sequences and define mathematically

$$T_{\min}(N, \phi) := \inf \text{costs}(N, \phi), \quad c_*(\phi) := \liminf_{N \rightarrow \infty} \frac{T_{\min}(N, \phi)}{N}. \quad (7)$$

Theorem 2 (First rung, direct Taylor bound, paper-level). *For every $0 < \phi \leq \pi$ and every order- N admissible sequence,*

$$T \geq 2 \left[2(N+1)! \sin\left(\frac{\phi}{4}\right) \right]^{1/(N+1)}. \quad (8)$$

Using Stirling's formula, $(N+1)!^{1/(N+1)} = (N+1)e^{-1}(1+o(1))$, and the fixed factor $[2 \sin(\phi/4)]^{1/(N+1)}$ tends to one. Hence

$$c_*(\phi) \geq \frac{2}{e}. \quad (9)$$

This rung is paper-level; it is pointwise weaker than Theorem 3 at every order and is omitted from the maximum in Corollary 6.

Theorem 3 (Elementary bound, valid at every order). *For every $0 < \phi \leq \pi$ and every order- N admissible sequence,*

$$T \geq 2 \left[2(2N+2)! \sin^2\left(\frac{\phi}{4}\right) \right]^{1/(2N+2)}. \quad (10)$$

*In particular, at $\phi = \pi$, $T \geq 2[(2N+2)!]^{1/(2N+2)}$, which exceeds $4(N+1)/e$ at every order; the limiting coefficient of N is $4/e \approx 1.4715$. This bound holds at every order with no threshold, no approximation theory, and no transcendental input beyond the factorial and a q -th root. It is machine-checked as **elementary_bound** (see Sec. VIII).*

Theorem 4 (Jensen bound, valid at every order). *Write $q = 2N+2$ and $s = 2 \sin^2(\phi/4)$. For every $0 < \phi \leq \pi$ and every order- N admissible sequence,*

$$T \geq \frac{\pi q}{e(2/s)^{1/q}}. \quad (11)$$

In particular, at $\phi = \pi$, $T \geq \pi q / (e 2^{1/q})$; the limiting coefficient of N is $2\pi/e \approx 2.3115$. The bound is explicit at every order, with deficit only $(2/s)^{1/q} - 1$ vanishing as $q \rightarrow \infty$. It is machine-checked as `jensen_rung_final` (and `jensen_rung_pi_final` at $\phi = \pi$); the per- N bound divided by N tends to $2\pi/e$ (Lean `jensen_ratio_tendsto`).

Theorem 5 (Exact-criterion rung, valid at every order). *Let $N \geq 1$, $q = 2N + 2$, $0 < \phi \leq \pi$, and $1 \leq \tau < q$. With the explicit closed-form quantities $\text{nrCcal}(\tau, q)$ and $\text{nrE2star}(\tau, q)$ of Appendix A, if*

$$\text{nrCcal}(\tau, q) q^2 \text{nrE2star}(\tau, q) < \sin^2\left(\frac{\phi}{4}\right), \quad (12)$$

then every order- N admissible sequence for the target ϕ satisfies

$$T > 2\tau. \quad (13)$$

The statement is machine-checked as `new_rung_final`: the hypothesis is the strict inequality $\tau < q$ with $1 \leq \tau$, and the conclusion is the strict bound $2\tau < \text{cost } \theta$. No threshold profile and no a -parameter enter: any τ passing the check certifies $T > 2\tau$.

Corollary 6 (A single bound at every order). *For every $N \geq 1$ and every $0 < \phi \leq \pi$, every order- N admissible sequence satisfies*

$$T \geq \max\left\{ 2 \left[2q! \sin^2(\phi/4) \right]^{1/q}, \pi q / (e(2/s)^{1/q}), 2\tau^* \right\}, \quad (14)$$

where $s = 2 \sin^2(\phi/4)$ and $\tau^ = \sup\{\tau \in [1, q) : \text{nrCcal}(\tau, q) q^2 \text{nrE2star}(\tau, q) < \sin^2(\phi/4)\}$, with the convention that the last term of the maximum is dropped when this set is empty. Consequently $c_*(\phi) \geq 4$, by combining the per-sequence bounds with the closed form (42), whose right-hand side is $(4 + o(1))N$; here $T_{\min}(N, \phi) = +\infty$ if no admissible sequence of order N exists. The Lean developments state the three per-sequence bounds in (14) (the paper-level $2/e$ first rung of Theorem 2 is pointwise weaker than Theorem 3 and is omitted from the maximum), and the passage to c_* is elementary analysis. All rungs used here cover $0 < \phi \leq \pi$; only the certified finite- N table below is stated at $\phi = \pi$.*

Table I compares the four rungs at sample orders. The Jensen rung is the strongest of them at every tabulated order (the table starts at $N = 2$; at $N = 1$ the elementary bound is larger), and the exact-criterion row takes over asymptotically, where its coefficient 4 exceeds $2\pi/e$. The ratio of the Jensen row to the elementary row tends to $\pi/2$, and the ratio of

TABLE I. The four rungs of the ladder—Theorems 2, 3, 4 and 5—at $\phi = \pi$ and sample orders N , with $q = 2N + 2$. Rows: direct, $2[(N + 1)! \sqrt{2}]^{1/(N+1)}$ (paper-level); elementary, $2(q!)^{1/q}$ (`elementary_bound_pi`); Jensen, $\pi q/(e^{1/q})$ (`jensen_rung_pi_final`); last row, the integers 2τ certified in Lean (`cert_N2–cert_N4`, `cert_N5_tight–cert_N12_tight`). Every entry is a lower bound on T ; the last row is strict, $T > 2\tau$, the other three are not. Bold marks the strongest proved lower bound at each order.

N	2	3	4	5	6	8	10	12
direct (paper-level)	4.08	4.83	5.58	6.34	7.10	8.62	10.13	11.64
elementary	5.99	7.53	9.06	10.58	12.09	15.11	18.11	21.10
Jensen	6.18	8.48	10.78	13.09	15.40	20.02	24.64	29.26
exact-criterion cert	2	2	4	8	10	16	22	28

the exact-criterion row to the Jensen row to $2e/\pi$. The first three rows evaluate the closed forms (8)–(11), whereas the last row consists of integers proved inside Lean (Sec. VI), not of evaluations.

IV. THE CARRIER AND THE FIRST TWO RUNGS

This section establishes the two cheapest rungs of the ladder and, with them, the fact that high-order robustification cannot be sublinear in the order. Both rungs need only one scalar object, the carrier $h(\lambda) = 1 - \frac{1}{2} \text{Re Tr}(U_1^\dagger U_\lambda)$ of Sec. IV A: order- N matrix flatness at the nominal point makes h flat there to order $q = 2N + 2$, while the derivatives of h are bounded by powers of the bandwidth $\tau = T/2$. One Taylor step at the endpoint $\lambda = 0$, where the costly evolution is switched off, then gives the paper-level bound $T \geq 2[2(N + 1)! \sin(\phi/4)]^{1/(N+1)}$ (Sec. IV B) and, with the doubled flatness order, the bound $T \geq 2[2q! \sin^2(\phi/4)]^{1/q}$ (Sec. IV C). Their asymptotic coefficients differ by exactly the factor by which the flatness order doubles.

A. Carrier setup (once)

Set

$$\tau := \frac{T}{2}, \quad q := 2N + 2, \quad (15)$$

and introduce the carrier

$$h(\lambda) := 1 - \frac{1}{2} \operatorname{Re} \operatorname{Tr}(U_1^\dagger U_\lambda). \quad (16)$$

Normalized in this way, h carries everything the two rungs need: it measures the separation of the endpoints, its vanishing order at the nominal point is twice the order of the matrix flatness, and it is a finite exponential sum of bandwidth τ .

1. Finite exponential bandwidth

Because

$$Z(\lambda\theta) = e^{-i\lambda\theta/2} \frac{I + \sigma_z}{2} + e^{+i\lambda\theta/2} \frac{I - \sigma_z}{2},$$

expanding the product in (3) shows that every matrix element of U_λ is a finite sum of exponentials $e^{i\mu\lambda}$ with

$$\mu = \frac{1}{2} \sum_{j=1}^L s_j \theta_j, \quad s_j \in \{-1, +1\}. \quad (17)$$

Consequently,

$$h(\lambda) = \sum_m \gamma_m e^{i\mu_m \lambda}, \quad |\mu_m| \leq \tau. \quad (18)$$

Thus $T/2$ is the bandwidth of the carrier: no component of h oscillates faster than τ .

2. Bounds on the carrier

The relative propagator $V_\lambda := U_1^\dagger U_\lambda$ is unitary. Therefore $|\operatorname{Tr} V_\lambda / 2| \leq 1$, and hence

$$0 \leq h(\lambda) \leq 2. \quad (19)$$

At the nominal point $\lambda = 1$, $V_1 = I$. Write locally

$$V_\lambda = a(\lambda)I + i\mathbf{b}(\lambda) \cdot \boldsymbol{\sigma}, \quad a^2 + |\mathbf{b}|^2 = 1,$$

with $a(1) = 1$ and $\mathbf{b}(1) = 0$. Matrix flatness of U_λ through order N implies flatness of V_λ through order N , so $\mathbf{b}$ has a zero of order at least $N + 1$ at $\lambda = 1$. Since

$$h = 1 - a = \frac{1 - a^2}{1 + a} = \frac{|\mathbf{b}|^2}{1 + a},$$

and the denominator does not vanish near $\lambda = 1$, the vanishing order of h is twice that of $\mathbf{b}$, hence at least $2N + 2$. This doubling is what the second rung exploits.

Proposition 7 (Carrier flatness). *For an order- N admissible sequence,*

$$h^{(k)}(1) = 0, \quad k = 0, 1, \dots, q - 1, \quad q = 2N + 2. \quad (20)$$

In particular, the scalar flatness is a consequence of the matrix flatness and is not an additional assumption.

3. The endpoint at $\lambda = 0$

At $\lambda = 0$, all Z factors become the identity, and therefore

$$U_0 = X(A), \quad A := \sum_{j=1}^L \alpha_j. \quad (21)$$

The target endpoint is $U_1 = Z(\phi)$. A direct $SU(2)$ calculation gives

$$\text{Tr}[Z(\phi)^\dagger X(A)] = 2 \cos \frac{\phi}{2} \cos \frac{A}{2}. \quad (22)$$

Hence

$$h(0) = 1 - \cos \frac{\phi}{2} \cos \frac{A}{2}. \quad (23)$$

For $0 < \phi \leq \pi$, $\cos(\phi/2) \geq 0$, so

$$h(0) \geq 1 - \cos \frac{\phi}{2} = 2 \sin^2 \frac{\phi}{4} > 0. \quad (24)$$

The distinction matters: the sharper-looking identity $h(0) = 1 - \cos(\phi/2)$ is false in general, because the free X rotations leave the arbitrary factor $\cos(A/2)$.

There is also a useful geometric interpretation. For $U, V \in SU(2)$,

$$\|U - V\|^2 = 2 - \text{Tr}(U^\dagger V), \quad (25)$$

so, in the present case,

$$h(0) = \frac{1}{2} \|U_1 - U_0\|^2. \quad (26)$$

Thus the lower bound (24) says that the target is uniformly separated from the set of endpoint propagators obtainable by turning off the costly Z evolution.

B. Direct propagator estimate: 2/e

This is the first rung of the ladder. It is paper-level: one Taylor step without doubling. Differentiating a single Z factor gives

$$\left\| \frac{d^m}{d\lambda^m} Z(\lambda\theta_j) \right\| = \left(\frac{\theta_j}{2} \right)^m.$$

Applying the product rule to the whole sequence and using the unitarity of all undifferentiated factors gives the multinomial estimate

$$\|U_\lambda^{(k)}\| \leq \sum_{k_1+\dots+k_L=k} \frac{k!}{k_1! \dots k_L!} \prod_j \left(\frac{\theta_j}{2} \right)^{k_j} = \tau^k. \quad (27)$$

Taylor's theorem about $\lambda = 1$, using the first N vanishing derivatives, gives

$$\|U_0 - U_1\| \leq \frac{\tau^{N+1}}{(N+1)!}. \quad (28)$$

On the other hand, $U_0 = X(A)$ and $U_1 = Z(\phi)$, and the exact endpoint geometry gives

$$\|X(A) - Z(\phi)\| \geq 2 \sin \frac{\phi}{4}. \quad (29)$$

Together these prove Theorem 2, stated in Sec. III.

This already rules out sublinear robustification: arbitrarily high-order cancellation costs at least $\Omega(N)$, and that conclusion needs nothing beyond the product rule and one Taylor step. The bound is pointwise weaker than Theorem 3 at every order, so it is omitted from the maximum in Corollary 6.

C. Using the doubled flatness: 4/e

The gain comes from the doubled flatness order. From (27) and $|\text{Tr } M| \leq 2 \|M\|$, one obtains

$$|h^{(k)}(\lambda)| \leq \tau^k. \quad (30)$$

By Proposition 7, the first $q - 1$ derivatives of h vanish at 1, where $q = 2N + 2$. Taylor's theorem therefore yields

$$|h(0)| \leq \frac{\tau^q}{q!}. \quad (31)$$

Combining with (24) gives

$$T \geq 2 \left[2q! \sin^2 \left(\frac{\phi}{4} \right) \right]^{1/q}, \quad q = 2N + 2. \quad (32)$$

Since $(q!)^{1/q} \sim q/e$, more precisely $(q!)^{1/q} = (q/e)(1 + o(1))$,

$$c_*(\phi) \geq \frac{4}{e} \approx 1.4715. \quad (33)$$

The factor of two in the asymptotic coefficient comes entirely from the doubled flatness order $q = 2(N + 1)$; no new physical assumption enters. The one piece of information still unused is that h is not merely flat and bounded but band-limited. Sec. V converts the bandwidth into a further factor $\pi/2$, giving the coefficient $2\pi/e$, and Sec. VI uses the whole band rather than one Taylor remainder point to reach a coefficient approaching 4.

V. THE JENSEN RUNG

The carrier h of (16) is a finite exponential sum, hence the restriction to $\lambda \in \mathbb{R}$ of an entire function. Let H be its canonical complexification. Only two analytic facts about H are needed: a growth bound in the imaginary direction, and Jensen's formula on discs. Write $\tau = T/2$ and $s = 2 \sin^2(\phi/4)$, so that $h(0) \geq s > 0$ by (24).

A. Complex growth and zero counting

Let H be the canonical complexification of h . It is entire and satisfies the pointwise growth bound

$$\|H(z)\| \leq 2e^{\tau|\Im z|}, \quad z \in \mathbb{C}, \quad (34)$$

since each X factor is unitary and each Z factor has norm $e^{|y|\theta_j/2}$, so submultiplicativity gives the bound with no dependence on the number of segments. Averaging $\log |H|$ over the circle $|z| = r$, the indicator mean $\frac{1}{2\pi} \int_0^{2\pi} |\sin \theta| d\theta = 2/\pi$ gives

$$\bar{M}(r) := \frac{1}{2\pi} \int_0^{2\pi} \log |H(re^{i\theta})| d\theta \leq \log 2 + \frac{2}{\pi} \tau r, \quad r \geq 0, \quad (35)$$

This is the non-asymptotic form of the Phragmén–Lindelöf indicator bound: $|H|$ grows only in the imaginary direction, and since the circle average of $|\sin \theta|$ is $2/\pi$, most of that average comes from near the real axis, where the growth is negligible.

Jensen's formula on the disc $D(0, r)$ reads $\bar{M}(r) = \log h(0) + N(r)$, where $N(r) = \sum_{|a_k| < r} \log(r/|a_k|)$ is the weighted count of the zeros a_k of H inside the disc; here H is scalar-valued, so the growth bound $\|H(z)\| \leq 2e^{\tau|\Im z|}$ of (34) is a bound on $|H(z)|$. The

flatness of Proposition 7 is a q -fold complex zero at $\lambda = 1$, and every such zero with $|1| < r$ contributes $\log(r/1) > 0$, so for $r > 1$

$$N(r) \geq q \log r. \quad (36)$$

Dropping unknown zeros only decreases the right-hand side, and using fewer than the q known zeros is strictly weaker. The q zeros at $\lambda = 1$ lie strictly inside the discs $|z| = r$ with $r > 1$ used below, so no zero on the boundary is involved, and the optimizer below is interior.

B. Proof of Theorem 4

Proof. Combining (35)–(36) with $h(0) \geq s$ gives, for all $r > 1$,

$$q \log r \leq \log \frac{2}{s} + \frac{2}{\pi} \tau r, \quad \text{i.e.} \quad \tau \geq \frac{\pi}{2} \frac{q \log r - \log(2/s)}{r}. \quad (37)$$

The right-hand side is optimized in closed form: $\sup_{r>1} (q \log r - L)/r = q/(e e^{L/q})$, attained at the interior point $r^* = e(2/s)^{1/q} > 1$. Hence $\tau \geq (\pi/2) q/(e(2/s)^{1/q})$, i.e.

$$T = 2\tau \geq \frac{\pi q}{e(2/s)^{1/q}}, \quad (38)$$

which is (11). Since $(2/s)^{1/q} \rightarrow 1$ and $q/N \rightarrow 2$, $\liminf_{N \rightarrow \infty} T/N \geq 2\pi/e$. The full chain is machine-checked as `jensen_rung_final`. $\square$

Remark 8 (Where the constant comes from, and where this lens caps). The gain of $\pi/2$ over the max-modulus (Schwarz) variant – which gives $T \geq 2q/(e(2/s)^{1/q})$, the same $4/e$ value as the Taylor route but by a genuinely different argument – is exactly the indicator mean $2/\pi$: Jensen averages over the circle instead of taking the worst point. Within the disc-Jensen lens actually executed (discs at either center, max-modulus, three-circles, known-zero products), the constant $2\pi/e$ is the proved ceiling: the estimate chain has no other tunable quantity, and the supremum is attained. Beating $2\pi/e$ by arbitrary-domain formulas, or reaching 4 by fuller complex analysis, is open; the obstruction is the uncertainty principle of Sec. IX.

VI. THE EXACT-CRITERION RUNG

The Jensen rung uses only zero density. The exact-criterion rung uses the whole band: the q vanishing derivatives at $\lambda = 1$ become q vanishing moments, into which the truncated

Chebyshev expansion of $e^{-i\mu}$ is inserted, and a compactly supported Fourier multiplier turns the replacement into a kernel estimate. Every constant below is the one that Lean checks, and the definitions are stated in Lean’s notation.

A. Vanishing derivatives become vanishing moments

With $h(\lambda) = \sum_m \gamma_m e^{i\mu_m \lambda}$ and $|\mu_m| \leq \tau$, differentiation gives $h^{(k)}(1) = i^k \sum_m \gamma_m \mu_m^k e^{i\mu_m}$, so Proposition 7 is equivalent to the moment constraints (Lean `h_moments`)

$$\sum_m \gamma_m \mu_m^k e^{i\mu_m} = 0, \quad k = 0, 1, \dots, q-1. \quad (39)$$

If P is any polynomial of degree $< q$, the moments imply $\sum_m \gamma_m P(\mu_m) e^{i\mu_m} = 0$, and since $h(0) = \sum_m \gamma_m$,

$$h(0) = \sum_m \gamma_m [1 - P(\mu_m) e^{i\mu_m}]. \quad (40)$$

With $\Psi(\mu) := e^{-i\mu} - P(\mu)$, the bracket has modulus $|\Psi(\mu)|$. Thus $h(0)$ is small whenever $e^{-i\mu}$ is uniformly approximable on the band $[-\tau, \tau]$ by a polynomial of degree $< q$. The development takes P to be the Chebyshev truncation of the Jacobi–Anger expansion (see, e.g., Ref. [19]); no Bessel API is used, only the integral definition of the Fourier coefficients.

The explicit closed-form quantities `nrCcal`(τ, q) and `nrE2star`(τ, q) entering (12)—the contour-shift coefficient bound, the collar bookkeeping, and the smoothing-kernel estimate—are stated verbatim in Lean’s notation in Appendix A ((A1)–(A4) and (A5)).

B. Proof of Theorem 5

Proof. For an admissible sequence with $T \leq 2\tau$ all frequencies satisfy $|\mu_m| \leq T/2 \leq \tau$, so by the moment conditions (39), $h(0) = \sum_m \gamma_m \Psi(\mu_m) = \int h \kappa$ (Lean `nr_main_rung`), whence $|h(0)| \leq \|h\|_\infty \|\kappa\|_1 \leq 2 \text{nrCcal}(\tau, q) q^2 \text{nrE2star}(\tau, q)$. With the endpoint bound $h(0) \geq 2 \sin^2(\phi/4)$ of (24) this contradicts (12). The wrapper applying this to the truncation data is machine-checked as `new_rung_final`. $\square$

C. Certified finite-order instances

At $\phi = \pi$ one has $\sin^2(\phi/4) = 1/2$ (and $h(0) = 1$ for every A by (23), since $\cos(\pi/2) = 0$), and the criterion (12) is verified inside Lean at the following bandwidths by rigorous explicit

bounds (rational square-root enclosures and Mathlib logarithm bounds; no `sorry` or `axiom`). Each row is an integer 2τ with $T > 2\tau$ proved by the named theorem:

TABLE II. Integers certified in Lean at $\phi = \pi$ by the exact criterion: order N , $q = 2N + 2$, the certified bandwidth τ (certificates `cert_N2`–`cert_N4` and `cert_N5_tight`–`cert_N12_tight` in column order), and the resulting lower bound $2\tau \leq \text{cost } \theta$, which the wrapper `new_rung_final` upgrades to the strict bound $T > 2\tau$ of Theorem 5. The certified values are not claimed to be maximal in τ ; with the same definitions the criterion also passes at $\tau = 2$ for $N = 3$ and $\tau = 3$ for $N = 4$.

N	2	3	4	5	6	7	8	9	10	11	12
$q = 2N + 2$	6	8	10	12	14	16	18	20	22	24	26
τ	1	1	2	4	5	7	8	9	11	12	14
$T > 2\tau$	2	2	4	8	10	14	16	18	22	24	28

D. Asymptotic closed form: coefficient 4

The criterion (12) implies a closed form with honest constants. With

$$\text{nrU}(q) := 6 \left(\frac{\log q}{q} \right)^{2/3}, \quad (41)$$

the development proves (Lean `nr_closed_form`) the following. Fix any explicit lower bound s_0 with $0 < s_0 \leq \sin^2(\phi/4)$, for instance $s_0 = \sin^2(\phi/4)$; note that s_0 bounds $\sin^2(\phi/4)$, whereas the s of Secs. V and VI is $2 \sin^2(\phi/4)$. Then every order- N admissible sequence with $N \geq 500 + \lceil \exp((11 + |\log s_0|)/1.2) \rceil$ satisfies

$$T \geq 4(N + 1) \left(1 - \text{nrU}(q) \right), \quad q = 2N + 2. \quad (42)$$

At $\phi = \pi$ with $s_0 = 1/2$ this threshold is $N \geq 500 + \lceil \exp((11 + \log 2)/1.2) \rceil$. Since $\text{nrU}(q) \rightarrow 0$, the coefficient of N approaches 4, and $\liminf_{N \rightarrow \infty} T_{\min}(N, \phi)/N \geq 4$ follows by elementary analysis. This is the closed-form face of the same rung: the constant 6 and the threshold are conservative bookkeeping from the explicit majorants, not features of the control problem.

Remark 9 (History: the threshold-4 mechanism and the $8/e$ variant). An earlier mechanism proved the same coefficient 4 through a one-parameter threshold profile $\rho_a(q) =$

$1 - a(\log q/q)^{2/3}$ with $a \geq 2$ (instances $N \geq 5010$ at $a = 2.3$ and $N \geq 46000$ at $a = 2.2$). It is superseded by Theorem 5, which certifies each bandwidth directly with no a -parameter and no threshold profile. A weak-constant variant valid at every order with no threshold, $T \geq 4(q! \sin^4(\phi/4)/(10^8 q^4))^{1/q}$ (Lean `eight_over_e_bound` with $C = 10^8$), has limiting coefficient $8/e \approx 2.943$, which *exceeds* the Jensen coefficient $2\pi/e \approx 2.311$: because of its 10^8 constant it is weaker than the Jensen bound below $N \approx 73$ and stronger from there on, while the coefficient-4 closed form (42) exceeds it from $N \approx 250$ and is proved for $N \geq 17557$. The four rungs above are ordered by the argument each one uses rather than by their constants, and this variant is recorded here for that reason and for the intermediate range it covers, not as a fifth rung. Both items in this remark are kept for the record only, and neither is used in any other proof.

VII. NUMERICAL UPPER BOUNDS FOR THE HARDEST TARGET

Corollary 6 gives $c_*(\phi) \geq 4$ and leaves the upper end of the window open. The gap is quantitative and worth measuring, and we take $\phi = \pi$, where the endpoint separation is largest. (At the trivial endpoints no positive bound can hold: $\phi = 0$ is realized by the empty sequence, and $Z(2\pi) = -I$ by the free rotation $X(2\pi)$.) This is also the target realized in the Rydberg protocol [13], where each segment of free evolution is designed to accumulate the phase π .

Every number here is an upper bound for $T_{\min}(N, \pi)$; the status of the data is summarized at the end of the section.

For a fixed number L of Z segments, write the sequence in the form (3),

$$U_\lambda = X(\alpha_1)Z(\lambda\theta_1)X(\alpha_2)Z(\lambda\theta_2) \cdots X(\alpha_L)Z(\lambda\theta_L),$$

and consider the finite-dimensional problem

$$T_{\min}(N, \pi) \leq \min \left\{ \sum_{j=1}^L \theta_j \right\}, \quad \theta_j \geq 0, \quad U_1 = Z(\pi), \quad \left. \frac{d^k U_\lambda}{d\lambda^k} \right|_{\lambda=1} = 0, \quad k \leq N. \quad (43)$$

Every admissible angle list satisfies Definition 1, so its cost bounds $T_{\min}(N, \pi)$ from above (not $c_*(\pi)$, which no single order can bound). One known sequence per order has all of its nonzero Z angles equal to π and total cost $T = (2N + 1)\pi$: at $N = 1$, $\theta = (\pi, \pi, \pi)$ with $\alpha = (2\pi/3, -2\pi/3, 2\pi/3)$ gives $U_1 = Z(\pi)$ and $U'_1 = 0$ exactly. This is the transcription to

the present model of the composite sequences of Ref. [3], which uses $2N$ pulses of area π in addition to the target pulse of area π . The free angles α_j are otherwise unconstrained, and the local search improves on that cost at every order (Table III).

a. How the numbers were obtained and checked. The minimizations used a truncated-Taylor propagator in the error variable with a time-budget parametrization, pushed down by an adaptive march and warm-started by continuation in the segment number and in the order. Every solution reported in Table III was then re-verified from scratch, with the three conditions kept apart:

- (i) *Nominal identity.* The nominal propagator U_1 is rebuilt with an independent `scipy.linalg.expm` product with no Taylor algebra, and compared with $Z(\pi)$ up to the harmless global sign $-I = X(2\pi)$, which costs nothing: a sequence realizing $-Z(\pi)$ realizes $Z(\pi)$ after the free outer rotation $X(2\pi)$. The mismatch $\|U_1 \mp Z(\pi)\|$ is at most 9×10^{-10} for every order $N \leq 12$.
- (ii) *Flatness.* With $u_k = d^k U_\lambda / d\lambda^k$ at $\lambda = 1$ and the a priori bound $\|u_k\| \leq \tau^k$ of (27), the dimensionless residual $\rho_N := \max_{1 \leq k \leq N} \|u_k\| / \tau^k$ is at most 1.5×10^{-10} in every row of Table III (at $N = 12$ the largest $\|u_k\|$ is 1.1×10^3 , against $\tau^{12} \approx 2 \times 10^{18}$). The coefficients are evaluated twice, by truncated convolution in double precision and by a matrix series at 50 significant decimal digits, agreeing on ρ_N to better than 1%; the tabulated values come from the 50-digit run.
- (iii) *Order N .* The deviation $\delta(\varepsilon) = \arccos(|\langle U_{1+\varepsilon}, Z(\pi) \rangle|/2)$ with $\langle U, V \rangle = \text{Tr}(U^\dagger V)$ has local logarithmic slope rising towards $N + 1$ as ε decreases, saturating once δ reaches the round-off floor (e.g. the last three slopes are 1.98, 1.99, 2.00 at $N = 1$). The third slope entries at $N = 9$ –11 sit at the round-off floor (they are 4.43, 1.67, 0.70 against $N + 1 = 10, 11, 12$); they are platform-dependent and are not result digits, and no order is inferred from them. Where the slope is resolved, the cancellation is carried to order N and stops there.

b. What the numbers say. Fig. 1 shows the same data graphically. Write $T_{\text{num}}(N, \pi)$ for the cost of the sequence found at order N ; by construction $T_{\text{min}}(N, \pi) \leq T_{\text{num}}(N, \pi)$, and the two need not be equal. Over the range in which the search depth is uniform, $N \leq 8$, those

TABLE III. One admissible sequence per order at $\phi = \pi$, for $N \leq 12$: number L of costly segments, cost $T = \sum_j \theta_j$ in radians (so $T_{\min}(N, \pi) \leq T$; no row is claimed to be optimal or to be a certified minimum), ratio to the reference cost $(2N + 1)\pi$ of Ref. [3], and the independent re-verification described in the text—nominal mismatch, dimensionless flatness residual $\rho_N = \max_{k \leq N} \|u_k\|/\tau^k$ of (27) with $\tau = T/2$, and the last three local logarithmic slopes of $\delta(\varepsilon)$, to be read against $N + 1$. The angle lists are distributed with the data set of Ref. [20]; slope entries that sit at the round-off floor are not result digits.

N	L	T (rad)	T/N	$T/[(2N+1)\pi]$	$\ U_1 \mp Z(\pi)\ $	ρ_N	slopes of δ
1	17	8.1505	8.150	0.865	$2.0 \cdot 10^{-10}$	$4.0 \cdot 10^{-11}$	1.98, 1.99, 2.00
2	21	13.3466	6.673	0.850	$2.5 \cdot 10^{-10}$	$1.0 \cdot 10^{-10}$	2.96, 2.98, 3.00
3	25	18.5505	6.183	0.844	$2.5 \cdot 10^{-12}$	$2.7 \cdot 10^{-12}$	3.94, 3.97, 3.99
4	31	23.7878	5.947	0.841	$1.6 \cdot 10^{-11}$	$4.5 \cdot 10^{-12}$	4.91, 4.96, 4.99
5	38	29.0593	5.812	0.841	$3.3 \cdot 10^{-10}$	$4.0 \cdot 10^{-11}$	5.89, 5.95, 5.99
6	48	34.2512	5.709	0.839	$3.3 \cdot 10^{-12}$	$1.4 \cdot 10^{-12}$	6.87, 6.94, 6.98
7	55	39.7030	5.672	0.843	$1.2 \cdot 10^{-10}$	$1.3 \cdot 10^{-11}$	7.84, 7.92, 7.93
8	58	44.9604	5.620	0.842	$4.3 \cdot 10^{-10}$	$3.4 \cdot 10^{-11}$	8.82, 8.91, 7.99
9	63	52.2657	5.807	0.876	$1.9 \cdot 10^{-11}$	$2.8 \cdot 10^{-12}$	9.63, 9.74, 4.43
10	70	56.9380	5.694	0.863	$7.0 \cdot 10^{-11}$	$1.2 \cdot 10^{-11}$	10.76, 10.58, 1.67
11	75	66.9068	6.082	0.926	$7.7 \cdot 10^{-10}$	$1.4 \cdot 10^{-10}$	11.66, 7.59, 0.70
12	78	67.6302	5.636	0.861	$8.7 \cdot 10^{-10}$	$7.8 \cdot 10^{-11}$	12.25, 12.51, 12.68

upper bounds increase by 5.2 ± 0.3 per order,

$$T_{\text{num}}(N, \pi) \approx 5.26 N + 2.81,$$

with increments 5.20, 5.20, 5.24, 5.27, 5.19, 5.45, 5.26 and residuals below 0.119. Every ratio $T/[(2N + 1)\pi]$ lies between 0.839 and 0.926, and every ratio T/N stays above the proved coefficient 4: it exceeds 2π at $N = 1, 2$ and falls below it from $N = 3$ on, reaching its smallest value 5.62 at $N = 8$. Part of the gain over that sequence may come from the free angles alone—that family fixes every α_j at π for algebraic convenience, whereas (43) leaves them free—but how much of it is available in this way is a question these numbers do not answer.

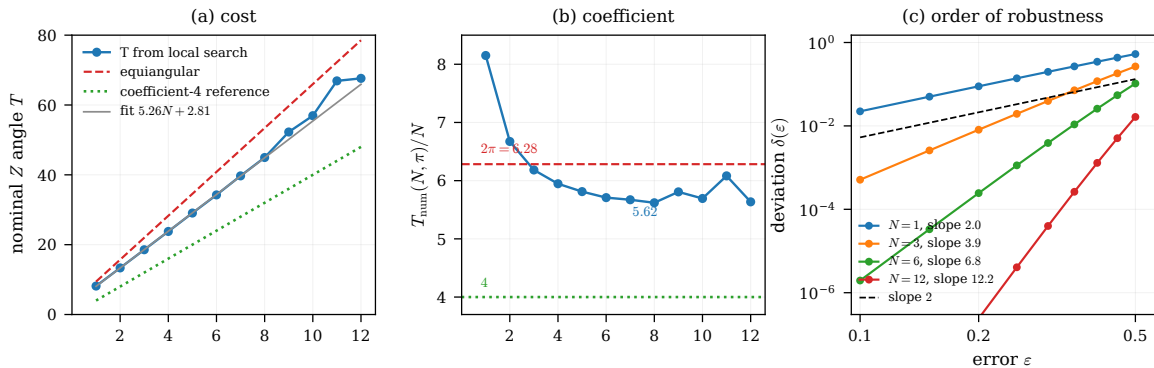


FIG. 1. The same data as Table III. (a) Cost found by the local search, against the reference cost $(2N + 1)\pi$ and the coefficient-4 line $4N$; the straight line is the fit $T \approx 5.26N + 2.81$ to the branch $N \leq 8$, whose residuals are below 0.12. (b) The same cost divided by N ; the horizontal references are the proved lower bound 4 and $2\pi \approx 6.28$, which the values at $N = 1, 2$ exceed. (c) The deviation $\delta(\varepsilon)$ on a log-log grid for $N = 1, 3, 6, 12$, with its local slope; the dashed line has slope 2, the smallest order appearing in the figure, and each curve is drawn only while δ stays above the round-off floor.

c. Status of the numbers, and what they do not claim. The optimization is a local search with no certificate of optimality, and what Sec. VII verifies is *only* that each printed sequence is admissible. Every entry of Table III is therefore an upper bound for $T_{\min}(N, \pi)$, and nothing more. In particular:

- The values are *not* claimed to be optimal. A cheaper sequence of the same order may exist, and $T_{\min}(N, \pi)$ may be strictly smaller than every number listed; these numbers give no lower bound at any finite order.
- No asymptotic claim is made. The fit $T \approx 5.26N + 2.81$ describes eight upper bounds with residuals below 0.119; it is not an estimate of $c_*(\pi)$, which these numbers cannot bound from either side. Because the quality of the search is itself N -dependent—the rows $N \geq 9$ come from a different seeding and are visibly looser—the increments of Table III cannot be extrapolated in N .
- The segment numbers L are those found by the search, not the smallest possible ones, and the comparison with that reference sequence is made at equal order N only.

None of this section enters the proofs of Theorems 3, 4 and 5. The one statement the data do establish is independent of optimality: the reference cost $(2N + 1)\pi$ is not optimal

at any order considered, since every listed cost is strictly below it, by a factor between 0.839 and 0.926. Closing the gap $4 \leq c_*(\pi) \leq 2\pi$ therefore requires either a construction in the free- X model that beats that construction, or a proof that no construction attains 4.

VIII. MACHINE-CHECKED VERIFICATION IN LEAN 4

Each rung of this paper corresponds to a named Lean 4 declaration of a public development [16]. The development consists of three independent trees: the analysis core and the exact-criterion rung live in `wC`, the Jensen rung in `wF`, and the weak-constant $8/e$ variant in `wM` (paths below are relative to `lean_sharp/`). In each tree the model – the rotations, the evolution, `Admissible`, and the cost $T = \text{cost } \theta$ – is fixed in `RobustZ/Statement.lean`, and the scalar flatness (20) is derived from the matrix flatness (`h_flat`) rather than assumed. Write $q = 2N + 2$ and $\tau = T/2$.

A. The machine-checked statements

Each rung corresponds to a declaration of the development, stated here with its exact hypotheses so that it can be compared with Table IV row by row. The elementary rung is `elementary_bound`, with `elementary_bound_pi` for the case $\phi = \pi$: for every $0 < \phi \leq \pi$ and every order- N admissible sequence, $T \geq 2(2q! \sin^2(\phi/4))^{1/q}$, and $T \geq 2(q!)^{1/q}$ at $\phi = \pi$. The Jensen rung is `jensen_rung_final`, with `jensen_rung_pi_final` at $\phi = \pi$: with $s = 2\sin^2(\phi/4)$, $T \geq \pi q/[e(2/s)^{1/q}]$, and `jensen_ratio_tendsto` records that this per-order bound divided by N tends to $2\pi/e$. The exact criterion is `new_rung_final`: for $N \geq 1$, $0 < \phi \leq \pi$ and $1 \leq \tau < q = 2N + 2$, the inequality $\text{nrCcal}(\tau, q) q^2 \text{nrE2star}(\tau, q) < \sin^2(\phi/4)$, with the closed forms (A1)–(A4), implies $2\tau < \text{cost } \theta$. The certified instances `cert_N2–cert_N4` and `cert_N5_tight–cert_N12_tight` (Table II) and the closed form $T \geq 4(N + 1)(1 - \text{nrU}(q))$ (`nr_closed_form`, i.e. (42)) are proved from it with nothing else added.

TABLE IV. Correspondence between the objects of this paper and the declarations that check them. File paths are relative to `lean_sharp/`; declaration names are exact.

Paper object	Lean declaration	File
Model, <code>Admissible</code> and <code>cost</code>	<code>Admissible, cost</code>	<code>wC/RobustZ/Statement.lean</code> (same in <code>wF</code> , <code>wM</code>)
Direct bound (8), Theorem 2	paper-level proof (in text)	Sec. IV
Carrier flatness (20), moments (39)	<code>h_flat, h_moments</code>	<code>wC/RobustZ/Flatness.lean</code> , <code>wC/RobustZ/ExpSum.lean</code>
Elementary bound, Theorem 3	<code>elementary_bound</code> , <code>elementary_bound_pi</code>	<code>wC/RobustZ/ElementaryBound.lean</code>
Jensen bound, Theorem 4, with its analytic core	<code>jensen_rung_final</code> , <code>jensen_rung_pi_final</code> , <code>jensen_ratio_tendsto</code>	<code>wF/RobustZ/JensenBridge.lean</code> , <code>wF/RobustZ/JensenRung.lean</code>
Coefficient bound L1, tail sum L2	<code>fcoef_sharp, besseL_L1, besseL_L2</code>	<code>wC/RobustZ/BesseLTail.lean</code>
Realization L3, Fourier inversion	<code>kap_L1_le, realize_L1</code> , <code>inversion_on_band</code>	<code>wC/RobustZ/RealizeLemma.lean</code> , <code>wC/RobustZ/InversionId.lean</code>
Chebyshev derivative bundle D1	<code>cb_term1_le, cb_term2_le</code> , <code>cb_hasSum0-cb_hasSum2</code>	<code>wC/RobustZ/ChebBundle.lean</code>
Exact criterion L4/L5 and closed form, Theorem 5	<code>nr_sup_le, nr_D1_le, nr_D2_le</code> , <code>nr_kappa_le, nr_main_rung</code> , <code>nr_closed_form</code>	<code>wC/RobustZ/NewRung.lean</code>
Wrapper and finite- N certificates	<code>new_rung_final, cert_N2-cert_N4</code> , <code>cert_N5_tight-cert_N12_tight</code>	<code>wC/RobustZ/RungFinal.lean</code> , <code>wC/RobustZ/RungTight.lean</code>
Weak-constant $8/e$ variant	<code>eight_over_e_bound</code>	<code>wM/RobustZ/EightOverE.lean</code>

B. Statement-by-statement correspondence

Table IV lists the correspondence between the objects of this paper and the declarations that check them.

C. Scope of the verification

Machine-checked: the model and the carrier flatness; the elementary bound with its exact constant 2, at every order and every $0 < \phi \leq \pi$; the Jensen bound with its exact constant π/e at every order and every $0 < \phi \leq \pi$; the exact criterion with its closed forms, hence every entry of Table II; the $8/e$ variant with $C = 10^8$; and the passage to the coefficient, as follows. `c_ge_four` proves $4 \leq \liminf_N T_{\min}(N, \phi)/N$ under two side conditions, that an admissible sequence exists at every order and that $T_{\min}(N, \phi)/N$ is eventually bounded above, and the dichotomy `c_ge_four_or_grows` keeps only the first and concludes either that same bound or that $T_{\min}(N, \phi)/N$ diverges to $+\infty$. The closed form (42) is `nr_closed_form`, whose hypothesis is the existence of truncation data at the bandwidth that form prescribes (constructed by `nrCheb_of_truncation`), rather than an unconditional per-sequence statement. Not machine-checked: the numerical material of Sec. VII. The low-order sequences found by local search and their re-verified costs are upper bounds computed outside Lean; they enter no proof, and no optimality or small- N statement about them is formalized. The audit ran on the development sources: a source scan finds no `sorry`, `admit`, or `axiom` in the project files, and the headline theorems depend only on the three standard logical axioms `propext`, `Classical.choice` and `Quot.sound` (see the `Audit*.lean` files, e.g. `wC/RobustZ/AuditRungFinal.lean`). Statements, build logs and the raw `#print axioms` output are collected at the project page, and the repository with the sources is cited as [16].

D. Code, data, and reproducibility

Everything used in this paper is public: the proofs are a Lean development, the numerical sequences and their independent re-verification are a data set with a browser front end, and the figures and tables of Sec. VII are regenerated from that data by short scripts. Table V

TABLE V. Public resources for this paper. The Lean development is self-contained: a fresh clone reproduces its statements with the three commands of its `README`. The numerical re-verification is likewise reproducible from the last entry, which carries the sequences at the precision used in Table III. All addresses were accessed on 8 October 2026.

Resource	Content
<code>https://show.dytchem.cn/bloch/</code>	Bloch-sphere animation and the full angle lists of every sequence of Table III; browser-side re-computation of the nominal mismatch, of the deviation $\delta(\varepsilon)$ and of its slope, with the reference sequence shown side by side. [20]
<code>https://show.dytchem.cn/lean/</code>	Verification status of the formal development: the exact Lean statements of the bounds, the source scan for <code>sorry</code> and <code>axiom</code> , and the raw <code>lakebuild</code> and <code>#print axioms</code> output for the headline theorems. [16]
GitHub repository	Lean 4 with Mathlib, pinned by <code>lean-toolchain</code> and <code>lake-manifest.json</code> ; <code>VERIFICATION.md</code> records two from-scratch verifications on three trees: <code>lean_sharp/wC</code> carries the analysis core, the exact-criterion rung and its certificates (54 modules); <code>lean_sharp/wF</code> carries the Jensen rung and its bridge (35 modules); <code>lean_sharp/wM</code> carries the 8/e variant (47 modules); all build green with no <code>sorry</code> or <code>axiom</code> in the sources. [16]
Verification scripts and data	<code>verify_num.py</code> , which rebuilds every propagator with <code>scipy.linalg.expm</code> and re-evaluates the Taylor coefficients of Sec. VII at 50 decimal digits; <code>make_outputs.py</code> , which regenerates Figs. 1 and 2 and the printable angle lists of Appendix B from the verified data; the data themselves (<code>num_results.json</code> , <code>solutions_largeN.json</code>); and a <code>README</code> . [20]

collects the four entry points.

Each tree builds green from its pinned toolchain and manifest alone. The angle lists of Appendix B are stored in the convention in which they were verified, rounded to 12 decimals of π ; the rounding is checked, not assumed: feeding those rounded lists back into (1)–(6)

again gives a nominal mismatch below 9×10^{-10} and a flatness residual below 1.5×10^{-10} , which are the two accuracy statements of Table III.

IX. DISCUSSION

The bounds are lower bounds on a resource, not a pulse-design prescription. Free and exact X rotations can rearrange how the systematic Z error accumulates, but when the common Z scale is turned down to $\lambda = 0$ every costly evolution disappears, and the carrier must connect two separated endpoints while being extremely flat at $\lambda = 1$.

A. What the ladder rules out and the orders that matter

At every order the cost is bounded below by the elementary bound, hence by $4(N+1)/e$: no arrangement of free rotations, however many, makes high-order robustification sublinear. The exact-criterion rung gives $c_*(\pi) \geq 4$, while the sequences of Ref. [3] transcribed to this model cost $T = (2N+1)\pi$ and hence give $c_*(\pi) \leq 2\pi$; so

$$4 \leq c_*(\pi) \leq 2\pi, \quad (44)$$

an interval of width $2\pi - 4 \approx 2.28$, whose endpoints differ by the factor $\pi/2$; the numerical costs of Sec. VII are fitted by $T \approx 5.26N + 2.81$ for $N \leq 8$ and lie within a factor 2.0 of the reference line $4N$ at $N = 1$ and within 1.4 at $N = 8$; the excess over $4N$ is bounded in relative terms, so the remaining question is one of a constant factor rather than of a new regime. Moreover, the model already grants the designer everything a fast-control layer could offer—exact, instantaneous and free X rotations in unlimited number and at arbitrary positions, with the target reached exactly at the nominal error—and the bound is still linear: here, a rich enough fast-control layer does not make the uncertain evolution asymptotically efficient.

a. Why the idealization makes the bounds conservative. The model is thin by design and every assumption favors the designer: one common multiplicative error, no time dependence, no decoherence, no leakage, instantaneous and perfect transverse control. Any model closer to a laboratory adds constraints, and adding constraints can only raise the least cost of an order- N robust sequence.

b. Finite order, incoherent errors, and the orders that matter. A device does not get to choose N freely: a longer sequence also spends more time in the presence of decay, the tension that shapes time-optimal gate design on Rydberg platforms [21]. Writing ω for the rate at which the uncertain Z -phase accumulates and γ for the decay rate, an order- N sequence of cost T pays an incoherent infidelity of order $\gamma T/\omega$ while its coherent error falls like ε^{N+1} , and the two balance at

$$N^* \approx \frac{\ln(\omega/\gamma)}{2 \ln(1/\varepsilon)}, \quad (45)$$

so the useful order grows only logarithmically as coherence improves. At the parameters of Fig. 2 the estimate gives $N^* \approx 3.0$, up to the order-one prefactor of the coherent term; the tabulated sequences place the minimum at $N^* = 2$.

This is where the finite-order form of the bounds matters: the relevant orders are small, and there the operative terms of (14) are the elementary and Jensen rungs, which supply the cost side of the balance at every order and without a threshold. The tradeoff cannot be evaded by inventing better pulses, only by improving γ , ε , or the resource model itself. Figure 2 evaluates the balance at parameters representative of current neutral-atom arrays: the optimum is at $N^* = 2$, and improving γ/ω by three orders of magnitude moves it only to 4.

B. Negatives: what cannot improve the ladder

a. No third algebraic doubling. The replacement method uses only two pieces of information: the band $|\mu_m| \leq \tau$ and the flatness $h^{(k)}(1) = 0$, $k < q$, with $q = 2N + 2$. The class of replacements available to the method is Π_{q-1} : every polynomial of degree $< q$ qualifies, and a continuous Q that annihilates the twisted moment measures of every admissible sequence lies in the uniform closure of Π_{q-1} on $[-\tau, \tau]$, which is Π_{q-1} itself because that space is finite-dimensional and hence closed. The converse richness statement — that the admissible moment measures are weak-* dense in the measures whose moments of order $< q$ vanish — is what would make the class exactly Π_{q-1} , and we do not prove it. Within this class, no determinant, character identity, or higher trace power adds a functional. Moreover, products preserve the flatness-to-band ratio: if h_1, h_2 have bands τ_1, τ_2 and flatness orders q_1, q_2 at 1, then $h_1 h_2$ has band $\tau_1 + \tau_2$ and flatness order $q_1 + q_2$, so the threshold $\tau < q$ — the

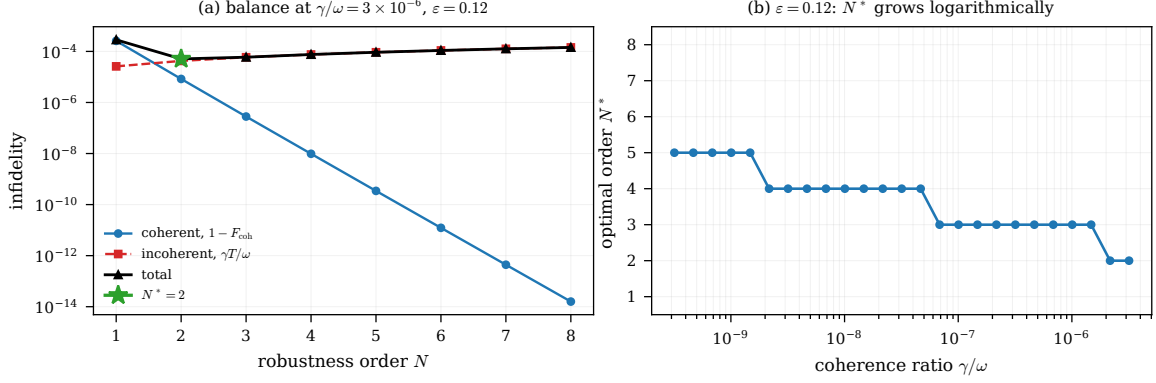


FIG. 2. The coherent-versus-incoherent balance at parameters representative of current neutral-atom arrays: interaction strengths $V/2\pi$ of a few hundred MHz at few-micrometre separations [14, 15] and Rydberg lifetimes of order $10^2 \mu\text{s}$ [22], which for $V/2\pi = 500$ MHz and $\gamma/2\pi \approx 1.6$ kHz give a coherence ratio $\gamma/\omega \approx 3 \times 10^{-6}$, together with a systematic variation $\varepsilon = 0.12$. (a) Coherent infidelity $\sin^2(\delta(\varepsilon)/2)$ of the sequences of Table III, the incoherent infidelity $\gamma T/\omega$ they accumulate, and the sum; the minimum is at $N^* = 2$. (b) The same balance over four decades of the coherence ratio; the optimum moves only to $N^* = 4$ after three decades and to $N^* = 5$ after four. The sequences plotted are those of Table III with $N \leq 8$; both terms are evaluated on those admissible sequences, so the plotted total is an upper bound on the infidelity achievable at that order.

source of the coefficient 4 – is unchanged by any power trick. The one explicit square-root variant is strictly worse because it halves the order: writing $V_\lambda = U_1^\dagger U_\lambda = a + i\mathbf{b} \cdot \boldsymbol{\sigma}$ gives $h = |\mathbf{b}|^2/(1+a)$, where each component b_j has band τ but flatness only $N+1$, so that route needs $\tau \lesssim N+1$, i.e. asymptotic coefficient 2. There is no third doubling of the coefficient, only a reparametrisation.

b. The minimax polynomial is marginal. Let $E_{q-1}(\tau)$ be the best uniform approximation error of $e^{-i\mu}$ on $[-\tau, \tau]$ by degree- $< q$ polynomials, and let ε^{tr} be the Chebyshev-truncation error used by the rung. The L^2 argument with the Chebyshev probability weight gives $\sqrt{2}|c_q| \leq E_{q-1}(\tau) \leq \varepsilon^{\text{tr}} \leq 2 \sum_{k \geq q} |c_k|$, so replacing the truncation by the optimal polynomial improves the error by at most a constant factor. Since the rung criterion is a condition on the error at the log level, a constant factor shifts the exponent by an additive constant only, and the asymptotic coefficient stays 4.

Remark 10 (Numeric illustration, not a theorem). Relaxing the error threshold by a factor $F = 2$ moves the largest admissible τ at $N = 12$ from 14.176 to 14.581, an increase of about

+2.9%; larger relaxations add a few further percent. These are computed illustrations only: the minimax direction is a refinement, never a new rung.

c. The free X -angle gives nothing at $\phi = \pi$. The exact endpoint identity is $h(0) = 1 - \cos(\phi/2)\cos(A/2)$ with $A = \sum_j \alpha_j$ free. At $\phi = \pi$ one has $\cos(\phi/2) = 0$, so $h(0) = 1$ for every A : no parity, topological, or other constraint on A can improve the endpoint bound in the worst case, which is the case the theorem is about. At $\phi < \pi$, $\min_A h(0) = 1 - \cos(\phi/2) = 2\sin^2(\phi/4)$, attained at $A \equiv \pi$, which is exactly the bound used here. The genuinely larger endpoint constant $|\mathbf{b}(0)| \geq \sin(\phi/2)$ belongs to the halved-order $\mathbf{b}$ -vector route above, which is asymptotically worse (its coefficient is 2, not 4), although at small ϕ and small N its larger endpoint constant can make it the stronger of the two.

C. Ceiling: the dichotomy at 4 and the open gap

Can any argument using only the band data (band τ , flatness order $q = 2N + 2$ at $\lambda = 1$, $\|h\|_\infty \leq 2$, endpoint $h(0) \geq 2\sin^2(\phi/4)$) certify $\liminf T/N > 4$? The answer is equivalent to one exact obstruction. Call $\text{UP}(q)$ the statement that there is a compactly supported density on $[-q, q]$ whose Fourier transform, while maximal at the origin, is as small as 2^{-q} times that maximum at $O(1)$ distances from it—equivalently, that the four band-data constraints at bandwidth q are consistent with an endpoint value $|h(0)| > 1$. Two implications fix the status of the ceiling. If $\text{UP}(q)$ holds for every q , then no band-data argument certifies a coefficient larger than 4: a witness at bandwidth $q = 2N + 2$ satisfies every hypothesis used at order N while realizing $T = 2q \approx 4N$, so no argument built from those hypotheses can exclude it. Conversely, if $\text{UP}(q)$ fails for every sufficiently large q , then the band data certify a coefficient larger than 4: at each such order the four constraints already exclude the corresponding bandwidth, so an argument passing 4 exists. Whether the ceiling at 4 holds is thus decided exactly when this uncertainty principle is decided for all large q , and this paper takes no position on it.

The ceiling at 4 is *open*. What is proved is the diagnosis on both sides. Every natural candidate witness family fails $\text{UP}(q)$ by the same quantitative mechanism – edge diffraction amplified by $|\cdot|^q$: sine powers give the 2π witness below but fail at 4; polynomial bumps, Gaussians with any cutoff width, and lattice profiles all fail with the endpoint-to-supremum ratio decaying exponentially in q . Nor is $\text{UP}(q)$ cheap to refute: no elementary argument

eliminates it. What is known unconditionally is threefold, and only the first of the three is proved in this paper: the rung with coefficient 4 (Theorem 5); a cap at 2π for every route that uses the band data through the replacement mechanism, argued here as follows — such a route needs the best uniform approximation error $\varepsilon_q(\tau)$ of $e^{-i\mu}$ on $[-\tau, \tau]$ by degree- $< q$ polynomials to fall below $s/2 \leq 1/2$, whereas $\varepsilon_q(\tau) \geq 1$ once $\tau \geq \pi q/2$ (if $|e^{-i\mu} - P| \leq \varepsilon$ then $|\cos \mu - \operatorname{Re} P| \leq \varepsilon$; $\cos$ alternates between ± 1 at $k\pi$, so for $\tau \geq \pi q/2$ there are more than q such points and Rolle forces $\operatorname{Re} P$ to be constant, whence $\varepsilon \geq 1$), with the extremal witness $g(x) = 2 \sin^q(\pi x/2)$ of type $\pi q/2$ attaining the endpoint 2; and the necessity that any admissible kernel have L^1 norm at least the best-approximation error, which is the mechanism behind Theorem 5 and is machine-checked as `realize_L1`. The 2π argument is a sketch in the sense that it bounds the mechanism of Secs. VI and not every conceivable band-data route. Thus $[4, 2\pi]$ is the exact open gap for band-data methods, sharp at the upper end by the sequence of Ref. [3], and this paper takes no position on whether the ceiling at 4 holds or fails.

a. Thresholds and numerics. The closed-form threshold of (42) is a sufficient condition assembled from explicit majorants — a contour-shift coefficient bound, a geometric tail sum, and kernel bookkeeping. The constant 6 and the threshold $500 + \lceil \exp((11 + |\log s_0|)/1.2) \rceil$ are conservative bookkeeping rather than features of the control problem. The criterion (12) being unverified below a certified bandwidth is therefore not evidence that the coefficient is smaller there; those orders are covered by the elementary and Jensen rungs.

D. Conclusions

The cost of cancelling a common systematic error to order N in the free- X model is linear in N , and the four rungs of this paper pin down how much of the coefficient is accessible to successively stronger arguments: $2/e$ and $4/e$ from one Taylor step, $2\pi/e$ once the carrier is treated as an entire function of exponential type, and a coefficient approaching 4 once its whole band is used. What is proved is therefore both a statement about the resource and a statement about the method. On the resource side the coefficient is linear in N and confined to $[4, 2\pi]$; on the method side, whether any argument that uses only the bandwidth, the flatness order, the supremum bound and the endpoint separation can pass 4 is equivalent to the uncertainty principle $\text{UP}(q)$ of Sec. IX C; the cap at 2π for the mechanism used here

is unconditional. At the small orders that a device with realistic coherence actually uses the elementary and Jensen rungs already supply the operative bound. The numerical sequences show that that sequence is not optimal at any order considered, so the upper end is not rigid either. Extending any of these statements—to non-common errors, to a full account of decoherence, or to a proof that the ceiling at 4 is attained or violated—would move the window, not the ladder.

ACKNOWLEDGMENTS

The author thanks the Lean 4 and Mathlib communities, whose libraries and tooling made the formal verification possible.

DATA AVAILABILITY

The Lean 4 formalization [16] and the numerical data set with its verification scripts [20] are openly available; both were accessed on 8 October 2026. The formal development builds from a fresh clone with its pinned toolchain and manifest, and the data set contains the angle lists behind Table III, the propagator re-verification, and the scripts that regenerate Figs. 1 and 2 and the tables of Sec. VII.

Appendix A: Exact-criterion rung: closed forms and kernel estimate

This appendix states verbatim, in Lean’s notation, the technical lemmas behind Theorem 5: the contour-shift coefficient bound, the collar bookkeeping, and the smoothing-kernel estimate. (The fourth technical ingredient, the flatness doubling of Proposition 7, is proved in Sec. IV A.) Nothing here adds a claim: every constant below is the one checked by Lean.

1. The closed-form quantities

For $\tau > 0$ and $k > \tau$ put

$$\alpha(\tau, k) := \operatorname{arcosh}(k/\tau), \quad f(\tau, k) := k \alpha(\tau, k) - \sqrt{k^2 - \tau^2}, \quad (\text{A1})$$

(Lean `besselAlpha`, `besselF`). The tail majorant and its closed-form sum are

$$B(k) := \sqrt{\frac{\pi}{8}} \frac{e^{-f(\tau,k)}}{(k^2 - \tau^2)^{1/4}}, \quad E_2^*(\tau, q) := \frac{2\sqrt{\pi/8} e^{-f(\tau,q)}}{\sqrt{\sqrt{q^2 - \tau^2} (1 - e^{-\alpha(\tau,q)})}}, \quad (\text{A2})$$

(Lean `majorB`, `nrE2star`), with $E_2(\tau, q) := 2 \sum_{m \geq 0} B(q + m) \leq E_2^*(\tau, q)$ (Lean `nrE2_le_E2star`, from `bessel_L2`). The weighted tail factors are, with $r = e^{-\alpha(\tau,q)}$,

$$\eta_1(r) := \frac{1+r}{(1-r)^3}, \quad \eta_2(r) := \frac{1+11r+11r^2+r^3}{(1-r)^5}, \quad (\text{A3})$$

(Lean `nrEta1`, `nrEta2`), and the collar width is $B_c := \tau/q^2$ (Lean `nrB`). Assembling the realization bound of the next subsection at this width gives the calibration factor

$$\text{nrCcal}(\tau, q) := \frac{2}{\pi} \frac{1}{q} \sqrt{2 \left(1 + \frac{B_c}{\tau}\right) \left(1 + \frac{4}{27} \eta_1\right) \left(32 \eta_2 q^2 + 44 + 2 \left(12 + \frac{16}{27}\right) \eta_1\right)}. \quad (\text{A4})$$

Here the square root covers the whole product and η_i abbreviates $\eta_i(e^{-\alpha(\tau,q)})$ (Lean `nrCcal`). The symbols τ and q are those of the paper, and $k > \tau$ is the tail index of (A1). The coefficient bound behind (A2) is the lemma `fcoef_sharp`; the tail sum is the lemma `bessel_L2`. The quantity $\text{nrU}(q)$ of Sec. VI is defined in (41), and nrE2 is E_2 below.

2. Realization and the kernel bound

Let $\Psi = e^{-i \cdot} - P$ with P the degree- $(q-1)$ Chebyshev truncation, and let χ be the explicit C^2 cutoff with $\chi = 1$ on the band, supported in the collar, and $\|\chi'\| \leq K/B_c$, $\|\chi''\| \leq K/B_c^2$ with $K = 4$. The development builds a kernel $\kappa \in L^1(\mathbb{R})$ whose Fourier transform agrees with Ψ on the band (Lean `inversion_on_band`) and controls its L^1 norm by the split-point optimization (Lean `kap_L1_le`, headline `realize_L1`):

$$\int_{\mathbb{R}} \|\kappa(t)\| dt \leq \text{nrCcal}(\tau, q) q^2 \text{nrE2}(\tau, q). \quad (\text{A5})$$

The inputs are the band error $\|\Psi\| \leq \text{nrE2}$ (Lean `nr_sup_le`) and the derivative bounds through the Chebyshev tail (Lean `nr_D1_le`, `nr_D2_le`, assembled in `nr_kappa_le`). With $\text{nrE2} \leq \text{nrE2star}$ this is the quantitative heart of the proof: no tunable parameter beyond τ and q enters, and the only condition on τ is the hypothesis $\tau < q$.

The scope distinctions are recorded in Sec. VIII: the elementary and Jensen rungs and the exact criterion are machine-checked end to end, while the passage from the per-sequence bounds to $c_*(\phi) \geq 4$ in Corollary 6 is ordinary analysis in the paper and no formalized infimum is claimed.

Appendix B: Numerical solutions

The angle lists behind Table III are distributed with the data set of Ref. [20], in the convention in which they were verified. For every order $N \leq 12$ at $\phi = \pi$ the data set gives the L costly angles θ_j and the L free angles α_j in radians (the printable listing divides them by π), both in the *time order* of the sequence: α_j is the free rotation that follows the costly segment θ_j , and the first entries are the ones applied first. The product that was verified is therefore

$$U_\lambda = X(\alpha_L)Z(\lambda\theta_L)X(\alpha_{L-1})Z(\lambda\theta_{L-1})\cdots X(\alpha_1)Z(\lambda\theta_1),$$

which is (3) with the indices relabeled, $j \mapsto L+1-j$. Two conventions turn the raw optimizer output into these lists, and both are exact rather than cosmetic: surplus free rotations, where the search produced $L+1$ of them, are merged with $X(a)X(b) = X(a+b)$, and every α_j is reduced to its representative in $[0, 2\pi)$, which uses $X(\alpha+2\pi) = -X(\alpha)$ and therefore changes the propagator only by the global sign that the convention $U_1 = Z(\pi)$ of (4) restores. After those two steps each row has L angles of each kind, as the sequence of the paper requires.

Each list was re-verified from scratch in exactly this form, as described in Sec. VII: the nominal mismatch $\|U_1 \mp Z(\pi)\|$ is below 9×10^{-10} and the flatness residual ρ_N of (27) is below 1.5×10^{-10} , so the lists carry the accuracy claimed in Table III, and the deviations from the target follow ε^{N+1} accordingly. The same lists, with the Bloch-sphere trajectory of every sequence and a browser-side re-computation of these numbers, are the first resource of Table V. The script that regenerates this appendix in printable form from the verified data is part of the same data set.

-
- [1] K. R. Brown, A. W. Harrow, and I. L. Chuang, Arbitrarily accurate composite pulse sequences, *Phys. Rev. A* **70**, 052318 (2004).
 - [2] J. T. Merrill and K. R. Brown, Progress in compensating pulse sequences for quantum computation, in *Quantum Information and Computation for Chemistry*, Advances in Chemical Physics, Vol. 154 (Wiley, 2014) pp. 241–294.
 - [3] G. H. Low, T. J. Yoder, and I. L. Chuang, Optimal arbitrarily accurate composite pulse sequences, *Phys. Rev. A* **89**, 022341 (2014).

- [4] S. Kukita, H. Kiya, and Y. Kondo, Lower bound on operation time of composite quantum gates robust against pulse length error, *J. Phys. A* **56**, 485305 (2023).
- [5] J. Zeng and X.-H. Deng, Fundamental costs of noise-robust quantum control: speed limits and complexity (2025), arXiv:2510.07183.
- [6] R. L. Kosut, D. A. Lidar, and H. Rabitz, A fundamental bound for robust quantum gate control (2025), arXiv:2507.01215.
- [7] C. Kabytayev, T. J. Green, K. Khodjasteh, M. J. Biercuk, L. Viola, and K. R. Brown, Robustness of composite pulses to time-dependent control noise, *Phys. Rev. A* **90**, 012316 (2014).
- [8] G. S. Uhrig, Keeping a quantum bit alive by optimized π -pulse sequences, *Phys. Rev. Lett.* **98**, 100504 (2007).
- [9] G. A. Paz-Silva and L. Viola, General transfer-function approach to noise filtering in open-loop quantum control, *Phys. Rev. Lett.* **113**, 250501 (2014).
- [10] G. S. Uhrig and D. A. Lidar, Rigorous bounds for optimal dynamical decoupling, *Phys. Rev. A* **82**, 012301 (2010).
- [11] K. Khodjasteh, T. Erdélyi, and L. Viola, Limits on preserving quantum coherence using multipulse control, *Phys. Rev. A* **83**, 020305 (2011).
- [12] H. K. Ng, D. A. Lidar, and J. Preskill, Combining dynamical decoupling with fault-tolerant quantum computation, *Phys. Rev. A* **84**, 012305 (2011).
- [13] B. Ye, D. Luo, W. Gong, *et al.*, Ultrafast high-fidelity Rydberg entangling gate (2026), unpublished manuscript, Massachusetts Institute of Technology; no public record at the time of writing.
- [14] S. J. Evered, D. Bluvstein, M. Kalinowski, S. Ebadi, T. Manovitz, H. Zhou, S. H. Li, A. A. Geim, T. T. Wang, N. Maskara, H. Levine, G. Semeghini, M. Greiner, V. Vuletić, and M. D. Lukin, High-fidelity parallel entangling gates on a neutral-atom quantum computer, *Nature* **622**, 268 (2023).
- [15] M. Mohan, R. de Keijzer, and S. Kokkelmans, Robust control and optimal Rydberg states for neutral atom two-qubit gates, *Phys. Rev. Research* **5**, 033052 (2023).
- [16] Dytchem, Explicit finite-order lower bounds for robust composite Z rotations: Lean 4 + Mathlib formalization, <https://github.com/Dytchem/robust-z-lower-bound-lean> (2026), verification status and raw audit output at <https://show.dytchem.cn/lean/>.

- [17] A. Bordg, H. Lachnitt, and Y. He, Certified quantum computation in Isabelle/HOL, *J. Autom. Reason.* **65**, 691 (2021).
- [18] A. Meiburg, L. A. Lessa, and R. R. Soldati, A formalization of the generalized quantum Stein’s lemma in Lean (2025), arXiv:2510.08672.
- [19] J. C. Mason and D. C. Handscomb, *Chebyshev Polynomials* (Chapman & Hall/CRC, Boca Raton, 2003).
- [20] Dytchem, Numerical upper bounds for high-order robust composite Z rotations: data set, verification scripts and browser front end, <https://show.dytchem.cn/bloch/> (2026), code and data at <https://show.dytchem.cn/files/code.zip>.
- [21] S. Jandura and G. Pupillo, Time-optimal two- and three-qubit gates for Rydberg atoms, *Quantum* **6**, 712 (2022).
- [22] I. I. Beterov, I. I. Ryabtsev, D. B. Tretyakov, and V. M. Entin, Quasiclassical calculations of blackbody-radiation-induced depopulation rates and effective lifetimes of Rydberg nS , nP , and nD alkali-metal atoms, *Phys. Rev. A* **79**, 052504 (2009), erratum: *Phys. Rev. A* **80**, 059902 (2009).